Marcus Kamuntu

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Summary

Final-year Computer Science (Data Science and AI) student and founder of Sendix, a SEND compliance platform for UK schools and councils. Building software for real clients since 15, across full-stack products, applied AI and security-conscious systems for regulated data.

Education

University of Dundee — Dundee, Scotland

BSc (Hons) Computer Science (Data Science and AI), First Class Honours expected — *Sep 2024 – Jul 2027*

- **Honours project:** attributing LLM-written student essays via reasoning structure, robust to "humaniser" tools (PyTorch).
- **Industrial project (NCR Atleos):** technical lead on an ATM diagnostics prototype that cleans logs from seven sources, scores anomalies with Isolation Forest and correlates them into incidents.
- VP, Dundee Computing Society: ran Quackathon 2026 (sold out, 80+ waitlist, NCR Atleos and BlackRock).
- Relevant coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Operating Systems, Multi-Paradigm Programming, Information & Network Security, Artificial Intelligence.

Experience

Sendix — Dundee, Scotland

Founder & Lead Engineer — *Jan 2026 – Present*

- Built and run a SEND compliance platform that checks EHCPs against the SEND Code of Practice and Scottish IEPs against Education Scotland guidance. Fixes the SENCO accepts go back to the council as **tracked changes in its own Word document**.
- FastAPI, PostgreSQL and Redis with a Next.js/TypeScript front end on Azure Container Apps; Llama 3.3 70B and Phi-4 run in Azure UK South with no prompt storage, and pupil names are redacted before analysis.
- Own security end to end: TOTP MFA, cross-tenant access-control tests, pupil data-rights tooling, CI security gates (Trivy, gitleaks, pip-audit), an approved DPIA and DPA, and Cyber Essentials.
- **£22.5k non-dilutive funding**: Scottish EDGE Young EDGE award (£15k), three awards at Dundee's Venture 2026, SEMTA 2026 finalist (Ethnic Minority Startup of the Year), and DOHE Go-Together accelerator, cohort 28.

Yarn Development Ltd — London, England (Remote)

Founder & Lead Developer — *Mar 2021 – Present*

- Founded at **15**; five years building education software for teachers and students, from first conversation to production. A client project for a maths teacher exposed the support-plan problem that became Sendix.
- Shipped a maths learning platform with a real-time-marked assessment engine and LaTeX rendering; QGenie, an LLM generator of GCSE/A-level exam-style questions; and ServStu, a Next.js platform with Yoti identity checks, OTP sign-in, payments and 91% test coverage.

Zamara Uganda — Kampala, Uganda

Pension Funds Summer Intern — *Aug 2024*

- Supported pension fund negotiations with institutional clients including Airtel Uganda and the Makerere University Retirement Benefits Scheme, working with senior stakeholders in a regulated financial environment.

Pollo Game Boat — Boston, MA, USA (Remote)

Co-Founder & Lead Developer — *Feb 2021 – Jun 2022*

- Scaled the Node.js/TypeScript backend to **250,000+ users**, migrating MySQL to MongoDB under load.

Projects

Unclaimed (**Winner, BlackRock track, GUTS 2026**) | *Python, FastAPI, PolicyEngine UK* — Oct 2026

- Tells a Glasgow household which three benefits to claim first and why. I built the calculation layer: a PolicyEngine UK adapter with a 2s budget inside a 5s deadline, and versioned rule packs.

Volunteer Driver Journeys (JPMorganChase Code for Good) | *TypeScript, React, Playwright* — Oct 2026

- For Glasgow Children's Hospital Charity, replacing spreadsheet coordination of volunteer drivers. I led the backend: one shared Zod contract for API and web, explicit state models for journeys, offers and notifications, Textmagic SMS fallback, and a deterministic seed plus Playwright script so every demo ran the same way.

FluxFile | *TypeScript, Next.js, BullMQ, Redis, Docker, Cloudflare R2* Feb 2026

- File converter: a BullMQ/Redis queue feeds containerised FFmpeg, LibreOffice and Sharp workers, so heavy jobs never block the web app.

Jotter (Build with Gemma hackathon, GDG Aberdeen) | *Next.js, Flask, Ollama, SQLite* Jul 2026

- Learning coach running Gemma 4 locally: one learner profile drives motivation nudges, diagnostic questions and grading of photographed handwritten work, with JSON-schema constrained model output validated by the app.

Local-Eval-Stack | *Python, FastAPI, llama.cpp, ChromaDB, MLflow, Docker* Apr 2026

- Self-hosted, OpenAI-compatible LLM gateway for constrained ARM hardware (4 vCPUs, 24 GB RAM), with PostgreSQL request logging, a RAG pipeline, and an evaluation harness that tracks LLM-as-judge scores per model in MLflow.

Web Analytics Data Warehouse | *Python, PostgreSQL, SQLAlchemy, Tableau* Feb 2026

- ETL pipeline that parses IIS server logs into a star schema, adds geolocation and crawler detection, and feeds Tableau dashboards on traffic quality, reach and response times.

Quartz-Lite | *Java, PostgreSQL* Dec 2025

- Job scheduler that persists jobs in PostgreSQL and uses SKIP LOCKED so concurrent workers never claim the same job.

Lecture Lens | *React Native, Expo, TypeScript, Supabase* Nov 2025

- Mobile study app that turns lecture PDFs, images and audio into summaries, spaced-repetition flashcards and Q&A, with row-level security on user data.

Autonomous EV Pathfinding | *Python, NetworkX, GeoPandas* Dec 2024

- A* routing over an OS Open Roads graph, weighing distance, traffic and energy use, with charging stations modelled as nodes.

Awards

Winner, BlackRock track, GUTS 2026 hackathon
Young EDGE award (£15k), Scottish EDGE, 2026
Three awards, University of Dundee Venture 2026
Finalist, Ethnic Minority Startup of the Year, SEMTA 2026
Junior Challenge Champion, DUCS Quackathon 2025

Skills

Languages: Python, TypeScript/JavaScript, SQL, Java, C, Bash
Domain: SEND and EHCPs (SEND Code of Practice), ASN and IEPs in Scotland, GDPR and DPIAs, selling to schools and councils
Frameworks & Data: FastAPI, Flask, Node.js, Express, React, Next.js, React Native, Prisma, PostgreSQL, Redis, MongoDB, SQLite, Supabase, Tableau
AI, Infra & Testing: LLM grounding, RAG, PyTorch, scikit-learn, pandas, llama.cpp, Ollama, ChromaDB, MLflow, Azure, Docker, GitHub Actions, pytest, Playwright